

Nomalungelo Mdleko

Practical 1: Basic SQL Syntax

Question 1

Databases

Worksheets

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Practical 2: SQL JOIN

Practical 1: SQL Fundamentals

COFFEE SHOP

ACCOUNTADMIN

COMPUTE_WH (X-Small)

Share

RETAIL_SHOP.SHOP_SALES

Settings

Open in Workspaces

Code Versions

1

2

3

4

5

6

7

--SELECT Statement

--Q1. Display all columns for all transactions

--Expected output: All columns

SELECT *

FROM RETAIL_SALES_DATASET;

Results

Chart

Search

Columns

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	#	TRANSACTION_ID	🕒	DATE	🔍	CUSTOMER_ID	🔍	GENDER	#	AGE	🔍	PRODUCT_CATEGORY	#	QUANTITY	#	PRICE
1		1		2023-11-24		CUST001		Male		34		Beauty		3		
2		2		2023-02-27		CUST002		Female		26		Clothing		2		
3		3		2023-01-13		CUST003		Male		50		Electronics		1		
4		4		2023-05-21		CUST004		Male		37		Clothing		1		
5		5		2023-05-06		CUST005		Male		30		Beauty		2		
6		6		2023-04-25		CUST006		Female		45		Beauty		1		
7		7		2023-03-13		CUST007		Male		46		Clothing		2		
8		8		2023-02-22		CUST008		Male		30		Electronics		4		
9		9		2023-12-13		CUST009		Male		63		Electronics		2		
10		10		2023-10-07		CUST010		Female		52		Clothing		4		
11		11		2023-02-14		CUST011		Male		23		Clothing		2		

Question 2

Databases

Worksheets

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Practical 2: SQL JOIN

Practical 1: SQL Fundamentals

COFFEE SHOP

ACCOUNTADMIN

COMPUTE_WH (X-Small)

Share

RETAIL_SHOP.SHOP_SALES

Settings

Open in Workspaces

Code Versions

6

7

8

9

10

11

12

FROM RETAIL_SALES_DATASET;

--Q2. Display only the Transaction ID, Date, and Customer ID for all records.

--Expected output: Transaction ID, Date, Customer ID

SELECT TRANSACTION_ID,

Date,

CUSTOMER_ID

Results

Chart

Search

Columns

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	#	TRANSACTION_ID	🕒	DATE	🔍	CUSTOMER_ID
1		1		2023-11-24		CUST001
2		2		2023-02-27		CUST002
3		3		2023-01-13		CUST003
4		4		2023-05-21		CUST004
5		5		2023-05-06		CUST005
6		6		2023-04-25		CUST006
7		7		2023-03-13		CUST007
8		8		2023-02-22		CUST008
9		9		2023-12-13		CUST009
10		10		2023-10-07		CUST010
11		11		2023-02-14		CUST011
12		12		2023-10-30		CUST012

Question 3

AccountADMIN - COMPUTE_WH (X-Small) Share

RETAIL_SHOP.SHOP_SALES Settings

Open in Workspaces Code Versions

```
14
15      --SELECT DISTINCT Statement
16      --Q3. Display all the distinct product categories in the dataset.
17      --Expected output: Product Category
18      SELECT DISTINCT PRODUCT_CATEGORY
19      FROM RETAIL_SALES_DATASET;
20
```

Results Chart

	PRODUCT_CATEGORY
1	Clothing
2	Beauty
3	Electronics

Question 4

AccountADMIN - COMPUTE_WH (X-Small) Share

RETAIL_SHOP.SHOP_SALES Settings

Open in Workspaces Code Versions

```
20
21      --Q4. Display all the distinct gender values in the dataset.
22      --Expected output: Gender
23      SELECT DISTINCT GENDER
24      FROM RETAIL_SALES_DATASET;
25
26      -----
```

Results Chart

	GENDER
1	Male
2	Female

Question 5

Practical 1: SQL Fundamen... Practical 2: SQL JOIN COFFEE SHOP +

AccountADMIN - COMPUTE_WH (X-Small) Share

RETAIL_SHOP.SHOP_SALES Settings

Open in Workspaces Code Versions

```
27
28      -- WHERE Clause
29      --Q5. Display all transactions where the Age is greater than 40.
30      --Expected output: All columns
31      SELECT *
32      FROM RETAIL_SALES_DATASET
33      WHERE AGE > 40 ;
```

Results Chart

	TRANSACTION_ID	DATE	CUSTOMER_ID	GENDER	AGE	PRODUCT_CATEGORY	QUANTITY	PRICE
1	3	2023-01-13	CUST003	Male	50	Electronics	1	
2	6	2023-04-25	CUST006	Female	45	Beauty	1	
3	7	2023-03-13	CUST007	Male	46	Clothing	2	
4	9	2023-12-13	CUST009	Male	63	Electronics	2	
5	10	2023-10-07	CUST010	Female	52	Clothing	4	
6	14	2023-01-17	CUST014	Male	64	Clothing	4	
7	15	2023-01-16	CUST015	Female	42	Electronics	4	
8	18	2023-04-30	CUST018	Female	47	Electronics	2	
9	19	2023-09-16	CUST019	Female	62	Clothing	2	
10	21	2023-01-14	CUST021	Female	50	Beauty	1	
11	24	2023-11-29	CUST024	Female	49	Clothing	1	

Question 6

The screenshot shows a SQL IDE interface. On the left, a sidebar lists 'Practical 1: SQL Fundamentals' and 'COFFEE SHOP'. The main editor displays a SQL query: `--06. Display all transactions where the Price per Unit is between 100 and 500. --Expected output: All columns SELECT* FROM RETAIL_SALES_DATASET WHERE PRICE_PER_UNIT BETWEEN 100 AND 500 ;`. Below the editor, the 'Results' tab is active, showing a table with 11 rows of transaction data.

	# TRANSACTION_ID	🕒 DATE	👤 CUSTOMER_ID	👤 GENDER	# AGE	📦 PRODUCT_CATEGORY	# QUANTITY	# PRICE
1	2	2023-02-27	CUST002	Female	26	Clothing	2	
2	4	2023-05-21	CUST004	Male	37	Clothing	1	
3	9	2023-12-13	CUST009	Male	63	Electronics	2	
4	13	2023-08-05	CUST013	Male	22	Electronics	3	
5	15	2023-01-16	CUST015	Female	42	Electronics	4	
6	16	2023-02-17	CUST016	Male	19	Clothing	3	
7	20	2023-11-05	CUST020	Male	22	Clothing	3	
8	21	2023-01-14	CUST021	Female	50	Beauty	1	
9	24	2023-11-29	CUST024	Female	49	Clothing	1	
10	26	2023-10-07	CUST026	Female	28	Electronics	2	
11	28	2023-04-23	CUST028	Female	43	Beauty	1	

Question 7

The screenshot shows a SQL IDE interface. On the left, a sidebar lists 'Practical 1: SQL Fundamentals' and 'COFFEE SHOP'. The main editor displays a SQL query: `--07. Display all transactions where the Product Category is either 'Beauty' or 'Electronics'. --Expected output: All columns SELECT* FROM RETAIL_SALES_DATASET WHERE PRODUCT_CATEGORY = 'Beauty' or PRODUCT_CATEGORY = 'Electronics' ;`. Below the editor, the 'Results' tab is active, showing a table with 11 rows of transaction data.

	# TRANSACTION_ID	🕒 DATE	👤 CUSTOMER_ID	👤 GENDER	# AGE	📦 PRODUCT_CATEGORY	# QUANTITY	# PRICE
1	1	2023-11-24	CUST001	Male	34	Beauty	3	
2	3	2023-01-13	CUST003	Male	50	Electronics	1	
3	5	2023-05-06	CUST005	Male	30	Beauty	2	
4	6	2023-04-25	CUST006	Female	45	Beauty	1	
5	8	2023-02-22	CUST008	Male	30	Electronics	4	
6	9	2023-12-13	CUST009	Male	63	Electronics	2	
7	12	2023-10-30	CUST012	Male	35	Beauty	3	
8	13	2023-08-05	CUST013	Male	22	Electronics	3	
9	15	2023-01-16	CUST015	Female	42	Electronics	4	
10	18	2023-04-30	CUST018	Female	47	Electronics	2	
11	21	2023-01-14	CUST021	Female	50	Beauty	1	

Question 8

RETAIL_SHOP.SHOP_SALES Settings Open in Workspaces Code Versions

```

46 --Q8. Display all transactions where the Product Category is not 'Clothing'.
47 --Expected output: All columns
48 SELECT*
49 FROM RETAIL_SALES_DATASET
50 WHERE PRODUCT_CATEGORY NOT IN ('Clothing');
51
52 --Q9. Display all transactions where the Quantity is greater than or equal to 3.

```

Results Chart

	# TRANSACTION_ID	🕒 DATE	👤 CUSTOMER_ID	👤 GENDER	# AGE	👤 PRODUCT_CATEGORY	# QUANTITY	# PRICE
1	1	2023-11-24	CUST001	Male	34	Beauty	3	
2	3	2023-01-13	CUST003	Male	50	Electronics	1	
3	5	2023-05-06	CUST005	Male	30	Beauty	2	
4	6	2023-04-25	CUST006	Female	45	Beauty	1	
5	8	2023-02-22	CUST008	Male	30	Electronics	4	
6	9	2023-12-13	CUST009	Male	63	Electronics	2	
7	12	2023-10-30	CUST012	Male	35	Beauty	3	
8	13	2023-08-05	CUST013	Male	22	Electronics	3	
9	15	2023-01-16	CUST015	Female	42	Electronics	4	
10	18	2023-04-30	CUST018	Female	47	Electronics	2	
11	21	2023-01-14	CUST021	Female	50	Beauty	1	

Question 9

ACCOUNTADMIN COMPUTE_WH (X-Small) Share

RETAIL_SHOP.SHOP_SALES Settings Open in Workspaces Code Versions

```

52 --Q9. Display all transactions where the Quantity is greater than or equal to 3.
53 --Expected output: All columns
54 SELECT*
55 FROM RETAIL_SALES_DATASET
56 WHERE QUANTITY >= 3;
57
58 -----

```

Results Chart

	# TRANSACTION_ID	🕒 DATE	👤 CUSTOMER_ID	👤 GENDER	# AGE	👤 PRODUCT_CATEGORY	# QUANTITY	# PRICE
1	1	2023-11-24	CUST001	Male	34	Beauty	3	
2	8	2023-02-22	CUST008	Male	30	Electronics	4	
3	10	2023-10-07	CUST010	Female	52	Clothing	4	
4	12	2023-10-30	CUST012	Male	35	Beauty	3	
5	13	2023-08-05	CUST013	Male	22	Electronics	3	
6	14	2023-01-17	CUST014	Male	64	Clothing	4	
7	15	2023-01-16	CUST015	Female	42	Electronics	4	
8	16	2023-02-17	CUST016	Male	19	Clothing	3	
9	17	2023-04-22	CUST017	Female	27	Clothing	4	
10	20	2023-11-05	CUST020	Male	22	Clothing	3	
11	23	2023-04-12	CUST023	Female	35	Clothing	4	

Question 10

The screenshot shows a SQL IDE interface. At the top, there's a header with 'RETAIL_SHOP.SHOP_SALES' and 'Settings'. Below the header is a code editor with the following SQL query:

```
58 -----  
59 -- Aggregate Functions  
60 --Q10. Count the total number of transactions.  
61 --Expected output: Total_Transactions  
62 SELECT COUNT(TRANSACTION_ID) AS TOTAL_TRANSACTIONS  
63 FROM RETAIL_SALES_DATASET;  
64
```

Below the code editor, there's a tab labeled 'Results' and a 'Chart' button. The results are displayed in a table with one column, '# TOTAL_TRANSACTIONS', and one row with the value 1000.

TOTAL_TRANSACTIONS
1000

Question 11

The screenshot shows a SQL IDE interface. At the top, there's a header with 'RETAIL_SHOP.SHOP_SALES' and 'Settings'. Below the header is a code editor with the following SQL query:

```
65 --Q11. Find the average Age of customers.  
66 --Expected output: Average_Age  
67 SELECT AVG(AGE) AS AVARAGE_AGE  
68 FROM RETAIL_SALES_DATASET;  
69  
70 --Q12. Find the total quantity of products sold.
```

Below the code editor, there's a tab labeled 'Results' and a 'Chart' button. The results are displayed in a table with one column, '# AVARAGE_AGE', and one row with the value 41.392000.

AVARAGE_AGE
41.392000

Question 12

The screenshot shows a SQL IDE interface. At the top, there's a header with 'RETAIL_SHOP.SHOP_SALES' and 'Settings'. Below the header is a code editor with the following SQL query:

```
70 --Q12. Find the total quantity of products sold.  
71 --Expected output: Total_Quantity  
72 SELECT SUM(QUANTITY) AS TOTAL_QUANTITY  
73 FROM RETAIL_SALES_DATASET;  
74  
75 --Q13. Find the maximum Total Amount spent in a single transaction.  
76 --Expected output: Max_Total_Amount
```

Below the code editor, there's a tab labeled 'Results' and a 'Chart' button. The results are displayed in a table with one column, '# TOTAL_QUANTITY', and one row with the value 2514.

TOTAL_QUANTITY
2514

Question 13

The screenshot shows a SQL IDE interface with a dark theme. At the top, there's a tab labeled 'COFFEE SHOP' and a user profile 'ACCOUNTADMIN' with a workspace 'COMPUTE_WH (X-Small)'. The main editor area contains a SQL query for Question 13. The query is as follows:

```
74 --Q13. Find the maximum Total Amount spent in a single transaction.
75 --Expected output: Max_Total_Amount
76 SELECT MAX(TOTAL_AMOUNT) AS MAX_TOTAL_AMOUNT
77 FROM RETAIL_SALES_DATASET;
78
79 --Q14. Find the minimum Price per Unit in the dataset.
80
```

Below the editor, the 'Results' tab is active, showing a single row of data:

	# MAX_TOTAL_AMOUNT
1	2000

Question 14

The screenshot shows the same SQL IDE interface. The main editor area contains a SQL query for Question 14. The query is as follows:

```
80 --Q14. Find the minimum Price per Unit in the dataset.
81 --Expected output: Min_Price_per_Unit
82 SELECT MIN(PRICE_PER_UNIT) AS MIN_PRICE_PER_UNIT
83 FROM RETAIL_SALES_DATASET;
84
85 -----
86 --GROUP BY Statement
```

Below the editor, the 'Results' tab is active, showing a single row of data:

	# MIN_PRICE_PER_UNIT
1	25

Question 15

RETAIL_SHOP.SHOP_SALES Settings

Open in Workspaces Code Versions

```

86      --GROUP BY Statement
87      --Q15. Find the number of transactions per Product Category.
88      --Expected output: Product Category, Transaction_Count
89      SELECT PRODUCT_CATEGORY, COUNT(TRANSACTION_ID) AS TRANSACTION_COUNT
90      FROM RETAIL_SALES_DATASET
91      GROUP BY PRODUCT_CATEGORY;

```

Results Chart

	PRODUCT_CATEGORY	TRANSACTION_COUNT
1	Beauty	307
2	Clothing	351
3	Electronics	342

Question 16

RETAIL_SHOP.SHOP_SALES Settings

Open in Workspaces Code Versions

```

92
93      --Q16. Find the total revenue (Total Amount) per gender.
94      --Expected output: Gender, Total_Revenue
95      SELECT GENDER, SUM(TOTAL_AMOUNT) AS TOTAL_REVENUE
96      FROM RETAIL_SALES_DATASET
97      GROUP BY GENDER;
98

```

Results Chart

	GENDER	TOTAL_REVENUE
1	Male	223160
2	Female	232840

Question 17

ACCOUNTADMIN COMPUTE_WF (X-Small) Share

RETAIL_SHOP.SHOP_SALES Settings Open in Workspaces Code Versions

```

--Q17. Find the average Price per Unit per product category.
--Expected output: Product Category, Average_Price
SELECT PRODUCT_CATEGORY, AVG(PRICE_PER_UNIT) AS AVARAGE_PRICE
FROM RETAIL_SALES_DATASET
GROUP BY PRODUCT_CATEGORY;

```

Results Chart

	PRODUCT_CATEGORY	AVARAGE_PRICE
1	Beauty	184.055375
2	Clothing	174.287749
3	Electronics	181.900585

Question 18

RETAIL_SHOP.SHOP_SALES Settings Open in Workspaces Code Versions

```

--Q18. Find the total revenue per product category where total revenue is greater than 10,000.
--Expected output: Product Category, Total_Revenue
SELECT PRODUCT_CATEGORY,
SUM(PRICE_PER_UNIT * QUANTITY) AS TOTAL_REVENUE
FROM RETAIL_SALES_DATASET
GROUP BY PRODUCT_CATEGORY;

```

Results Chart

	PRODUCT_CATEGORY	TOTAL_REVENUE
1	Beauty	143515
2	Clothing	155580
3	Electronics	156905

Question 19

RETAIL_SHOP.SHOP_SALES Settings Open in Workspaces Code Versions

```

--Q19. Find the average quantity per product category where the average is more than 2.
--Expected output: Product Category, Average_Quantity
SELECT PRODUCT_CATEGORY,
AVG(QUANTITY) AS AVARAGE_QUANTITY
FROM RETAIL_SALES_DATASET
GROUP BY PRODUCT_CATEGORY;

```

Results Chart

	PRODUCT_CATEGORY	AVARAGE_QUANTITY
1	Beauty	2.511401
2	Clothing	2.547009
3	Electronics	2.482456

Question 20

127 SELECT TRANSACTION_ID,
128 TOTAL_AMOUNT,
129 CASE
130 WHEN TOTAL_AMOUNT > 1000 THEN 'HIGH'
131 ELSE 'LOW'
132 END AS SPENDING_LEVEL
133 FROM RETAIL_SHOP_SALES_DATASET.

Results

Chart

Results

Chart

	# TRANSACTION_ID	# TOTAL_AMOUNT	A SPENDING_LEVEL
1	1	150	LOW
2	2	1000	LOW
3	3	30	LOW
4	4	500	LOW
5	5	100	LOW
6	6	30	LOW
7	7	50	LOW
8	8	100	LOW
9	9	600	LOW
10	10	200	LOW
11	11	100	LOW
12	12	75	LOW

Question 21

RETAIL_SHOP.SHOP_SALES Settings

Open in Workspaces Code Versions

127 SELECT TRANSACTION_ID,
128 TOTAL_AMOUNT,
129 CASE
130 WHEN TOTAL_AMOUNT > 1000 THEN 'HIGH'
131 ELSE 'LOW'
132 END AS SPENDING_LEVEL
133 FROM RETAIL_SHOP_SALES_DATASET.

Results

Chart

Results

Chart

	A CUSTOMER_ID	# AGE	A AGE_GROUP
1	CUST001	34	ADULT
2	CUST002	26	YOUNG
3	CUST003	50	ADULT
4	CUST004	37	ADULT
5	CUST005	30	ADULT
6	CUST006	45	ADULT
7	CUST007	46	ADULT
8	CUST008	30	ADULT
9	CUST009	63	SENIOR
10	CUST010	52	ADULT
11	CUST011	23	YOUNG
12	CUST012	35	ADULT

